

Chedam: work log

Cumulative record of every working day, newest first. This copy covers up to Monday, October 5, 2026. Owner: Sreya.
Baseline: Master Specification v1.0.

Days in this log

- **2026-10-05:** Hub base re-done on the new network, display made switchable, phones trust the hub, work log started
- **2026-10-03:** New network: own router, Pi re-flashed, SSH keys documented, HDMI fixed
- **2026-10-02:** P0 kickoff: spec review, dev PC tools, first Pi flash and first hub setup (TELUS network)

Hub base re-done on the new network, display made switchable, phones trust the hub, work log started

Summary

- Full health check of the re-flashed Pi: healthy, OS up to date, no hub software yet.
- Restored admin access (the sudoers rule is lost on every re-flash).
- Changed `setup-hub.sh` so the display setting can be switched **both ways** (monitor on or headless) and re-runs leave the boot config unchanged.
- Ran hub setup with `--keep-display`, rebooted and verified everything. HTTPS works by name and by IP.
- iPhone 13 Pro and Android 9 installed the new "Chedam Store CA" and opened the hub page without warnings ("Chedam hub is running").
- Started this cumulative work log (`Work Log/`).
- P0 step 1 (hub base) is complete.

Naming note: the local folder `chedam-updates/` in the project is Sreya's YouTube and notes repo (gitignored). It is **not** the update channel. Nothing is renamed; the public update-package repo name will be decided when step 9 (Updates) starts.

1. Health check (before any change)

One SSH session ran these checks:

```
whoami; hostname; hostname -I; cat /proc/device-tree/model; grep PRETTY /etc/os-release; uname -r
uptime; sudo -n true; free -m; df -h / /boot/firmware; lsblk
vcgencmd measure_temp; vcgencmd get_throttled; vcgencmd measure_volts core
timedatectl; ip route; ping -c2 1.1.1.1
apt list --upgradable; dpkg-query -W <packages>; systemctl is-active ...; systemctl --failed
cat /boot/firmware/config.txt /boot/firmware/cmdline.txt
```

Area	Result
Hardware	Pi Zero 2 W Rev 1.0, 33.2 °C, throttled 0x0, core 1.2875 V
OS	Debian 13 trixie, kernel 6.18.50+rpt-rpi-v8, 0 pending updates
Uptime	1 day 12 h, load 0.00, no failed units
Memory / disk	415 MB total, 273 MB available / 119 GB card, 2.3 GB used
Network	192.168.50.101 via 192.168.50.1, internet ping 7 ms
Time	NTP synced, America/Vancouver, no RTC
sudo	Needed a password (re-flash reset it)
Hub software	Not installed: no caddy, chrony, ufw, sqlite3 or <code>/opt/chedam</code>
Display	<code>video=HDMI-A-1:1280x720@60D</code> present (monitor in use)

2. Admin access

Failed case: `Permission denied (publickey)` when restoring sudo

- The `ssh -t chedam "...sudo tee..."` command was typed **inside an SSH session on the Pi** (prompt `chedam@chedam:~ $`).
- So the Pi tried to SSH into itself and had no key for that. Re-flashing was not needed.
- Fix: run only the inner command on the Pi:

```
echo 'chedam ALL=(ALL) NOPASSWD:ALL' | sudo tee /etc/sudoers.d/010-chedam && sudo chmod 440 /etc/sudo
sudo -n true && echo OK
```

3. Display becomes a switch (DL-24 updated)

Sreya's requirement: the monitor is optional on the Zero 2 W pilot, but the option must stay. Pi 4 production hubs will use a display to operate the hub directly.

Change in `hub/scripts/setup-hub.sh` (boot config section):

- Removes any earlier chedam block, then writes a block between `# chedam-begin` and `# chedam-end`.
- `--keep-display`: display driver `vc4-kms-v3d` and display detection **on**, Bluetooth off.
- No flag: headless. The display driver and detection are off, `gpu_mem=16`, Bluetooth off; this saves ~50 MB of RAM.
- Audio and camera detection are always off.

```
sudo bash ~/hub/scripts/setup-hub.sh --keep-display # monitor on (current)
sudo bash ~/hub/scripts/setup-hub.sh # headless
sudo reboot # after switching
```

Failed case: an extra `[all]` line on every run

- The test (switching off, on, off, on on a copy of `config.txt`) worked, but each run appended another `[all]` line. It was harmless but untidy.
- Fix: moved `[all]` inside the marked block. The original config (`config.txt.chedam-orig`, saved by the first run) was restored and setup run twice. Both runs gave the same checksum (`d91ff929`), and the only differences from the original were audio off, camera detection off and the chedam block.

4. Hub setup on 192.168.50.101

```
scp -r scripts system chedam:hub/
# on the Pi: download + verify PocketBase
wget pocketbase_0.40.4_linux_arm64.zip checksums.txt
grep linux_arm64 checksums.txt | sha256sum -c - # OK
sudo apt-get update
sudo bash ~/hub/scripts/setup-hub.sh --keep-display
sudo systemctl reboot
```

Verification after reboot:

Check	Result
Services	chedam-hub, caddy, chrony, avahi-daemon, ufw active; no failed units
Memory	PocketBase 28 MB + Caddy 37 MB = 65 MB (budget 150 MB, NFR-08); 264 MB available with display on
Watchdog	<code>RuntimeWatchdogUsec=14s</code>
Time	chrony, 0.0000007 s from NTP, leap normal
Temperature	38.6 °C, throttled 0x0
Display	<code>/dev/dri/card0</code> present
Firewall	22, 80, 443/tcp and 5353/udp allowed (IPv4 + IPv6)
CA download	<code>http://chedam.local/ca.crt</code> → 200, "Chedam Store CA - 2026 ECC Root", valid to 2036-08-13
HTTPS	<code>https://chedam.local/api/health</code> and <code>https://192.168.50.101/api/health</code> → 200
Certificate check	<code>openssl s_client ... -CAfile ca.crt</code> → Verify return code 0, with and without SNI
HTTP	Redirects to HTTPS (302)
Caddy errors since boot	0

PC test commands (Git Bash; `--resolve` avoids Windows mDNS problems, and `--ssl-no-revoke` avoids the Schannel revocation error):

```
curl --resolve chedam.local:80:192.168.50.101 -o ca.crt http://chedam.local/ca.crt
curl --ssl-no-revoke --resolve chedam.local:443:192.168.50.101 --cacert ca.crt https://chedam.local/a
curl --ssl-no-revoke --cacert ca.crt https://192.168.50.101/api/health
```

5. Phones (risk R5, first part)

Device	How	Result
iPhone 13 Pro, iOS 26.6.1	Safari <code>http://chedam.local/ca.crt</code> → install profile → Settings > General > About > Certificate Trust Settings → on	<code>https://chedam.local</code> shows the placeholder, no warning
Android 9	Chrome <code>http://192.168.50.101/ca.crt</code> → install for VPN and apps	<code>https://192.168.50.101</code> shows the placeholder, no warning

The camera test over HTTPS comes later, when the client app exists.

6. Work log started

- Folder `Work Log/`: one Markdown source per day in `Work Log/entries/YYYY-MM-DD.md`.
- `tools/worklog/build_worklog.py` builds `Work Log/Chedam-Work-Log-YYYY-MM-DD.pdf` with **all** days, newest first.
- Earlier PDFs are kept, so each day has its own complete snapshot.

```
python tools/worklog/build_worklog.py          # newest day
python tools/worklog/build_worklog.py 2026-10-05 # a specific day
```

The days 2026-10-02 and 2026-10-03 were written up from the session records and `docs/P0-pi-setup-log-2026-10-03.pdf`.

Next

P0 step 2: schema v1 and the event log (Section 10 tables for setup, people, roles, devices, storage areas, tasks, events, backups and updates; common fields on every table; an `events` row on every write; sample data first).

2026-10-03 (Saturday, October 3, 2026)

New network: own router, Pi re-flashed, SSH keys documented, HDMI fixed

Source: docs/P0-pi-setup-log-2026-10-03.pdf (detailed log of this day) and docs/P0-restart-new-network.md (the plan). This entry carries its content forward.

Summary

- The TELUS modem gives no admin access, so DHCP reservations are impossible. The dev hub moved behind our **own TP-Link router**, which is LAN-cabled to the bedroom wall port.
- Network: subnet `192.168.50.0/24`, gateway `192.168.50.1`, 2.4 GHz SSID `TP-LINK_2CBE`.
- Pi `chedam` at **192.168.50.101** (MAC `88:A2:9E:5C:17:65`, reservation enabled); dev PC DESKTOP-AHG4TD7 at `192.168.50.100`.
- The SD card was wiped and re-flashed. Key-only SSH works, and the OS is updated.
- HDMI console made visible. `setup-hub.sh` gained the `--keep-display` option.
- Commits: `ea0a7e6`, `678eee4`, `bacf585`.

Why SSH keys instead of a password

With a key pair, the private key never leaves the PC and the Pi holds only the public key. No secret crosses the network, keys cannot be guessed, and scripts can log in, so password login is switched off.

File	Where	Rule
<code>id_ed25519</code>	PC <code>C:\Users\Venkata\.ssh\</code>	Private. Never share, paste or commit; back it up privately
<code>id_ed25519.pub</code>	PC, same folder	Public, one line ending <code>chedam-dev</code> . Safe to copy
<code>authorized_keys</code>	Pi <code>/home/chedam/.ssh/</code>	Allowed public keys; folder 700, file 600
<code>known_hosts</code>	PC <code>.ssh\</code>	Remembered Pi identities; <code>ssh-keygen -R <ip></code> after a re-flash
<code>config</code>	PC <code>.ssh\</code>	Alias <code>ssh chedam</code>

Key procedures recorded:

- create the key pair
- put the public key on the Pi with Imager (REMOVE old keys, then BROWSE to the `.pub` file)
- add or replace keys with `type new.pub | ssh chedam "cat >> ~/.ssh/authorized_keys"`
- what to do when locked out (monitor + keyboard, or re-flash)

Steps

1. Backed up `.ssh` with `Copy-Item $env:USERPROFILE\.ssh $env:USERPROFILE\.ssh_backup -Recurse`.
2. Cleared old host fingerprints by deleting `known_hosts` and `known_hosts.old`. There was a backup.
3. Router: changed the admin password, set the LAN to `192.168.50.1/24` (different from TELUS `192.168.1.x`), kept it in router mode, and turned on 2.4 GHz.
4. Re-flashed the card. Imager: Pi Zero 2 W, Lite 64-bit, `chedam/chedam`, new Wi-Fi, CA, SSH public-key only.
5. Found the Pi in the router's DHCP client list and created the reservation.
6. Updated the SSH config to `HostName 192.168.50.101` and removed `HostKeyAlias`.
7. Ran `sudo apt update && sudo apt full-upgrade -y` and checked that `sshd -T` shows `passwordauthentication no`.
8. Readiness check: model, OS, IP, active Wi-Fi, `multi-user.target`; reboot test passed.
9. Forced HDMI output (see below).

Failed cases and fixes

Attempt / symptom	Cause	Fix
ssh-keygen -R <old-ip> errored	PowerShell treats < as an operator	Use real addresses; deleting known_hosts avoided needing the old IP
Key pair deleted during "start fresh"	Not needed: a key is not tied to a network	Restored from .ssh_backup
ssh 192.168.50.101 denied as venkata	No user given, so the Windows name was used	ssh chedam@192.168.50.101
ssh chedam@chedam timed out	Windows cannot resolve a bare name	IP, chedam.local, or the alias
-i \$env:venkata\.ssh\... not found	\$env:USERPROFILE is a real variable, not a placeholder	Keep \$env:USERPROFILE
Permission denied (publickey)	Imager wrote an old remembered key (... COL3o srey...), not the PC's (... LuhpI0dY chedam-dev)	Re-flash; REMOVE old keys, BROWSE to id_ed25519.pub
A command pasted into Imager's key box	Get-Content ... \ Set-Clipboard text pasted as the key	REMOVE, then BROWSE to the file
"REMOTE HOST IDENTIFICATION HAS CHANGED"	Expected after a re-flash (new host key)	ssh-keygen -R 192.168.50.101
Reservation saved as Disabled (TP-Link)	Router default	Tick it, then Enable Selected; leave ARP binding off
HDMI blank	dmesg: vc4-drm: Cannot find any crtc or sizes	Append video=HDMI-A-1:1280x720@60D to the single line in /boot/firmware/cmdline.txt, then reboot
Lines after sudo cp lost	The sudo password prompt ate the pasted lines	Paste one command at a time when sudo may prompt

HDMI fix commands:

```
sudo cp /boot/firmware/cmdline.txt /boot/firmware/cmdline.txt.bak
sudo sed -i '1 s/$/ video=HDMI-A-1:1280x720@60D/' /boot/firmware/cmdline.txt
sudo reboot
```

Lessons

- Before flashing, check that the key shown in Imager ends the same as type C: \Users\Venkata\.ssh\id_ed25519.pub. Imager remembers old keys.
- Always back up .ssh before changes.
- With password login off, losing id_ed25519 means re-flashing or using a keyboard and screen.
- No passwords or key contents are stored in the repo.

P0 kickoff: spec review, dev PC tools, first Pi flash and first hub setup (TELUS network)

Summary

- Read Master Specification v1.0 and pulled out the P0 (Foundation) scope, done-criteria and phase gate.
- Wrote `docs/P0-readiness-checklist.md`: hardware, software with versions and links, Pi flashing steps, accounts.
- Installed and verified the developer tools on the Windows PC (all checksums matched the official ones).
- Flashed the Pi. The first attempts failed on SSH keys (including the private key being pasted into Imager); the key was replaced and SSH worked.
- Wrote the hub setup (`hub/scripts/setup-hub.sh` + `hub/system/*`) and ran it on the Pi at `192.168.1.241` (TELUS network). HTTPS worked by name and IP.
- Wrote `docs/P0-mini-spec.md` (P0 build order, decisions DL-21 to DL-25).
- Decisions from Sreya: RTC skipped for now; cloud backup tested later when client IDs exist; test phones are Android 9 and iPhone 13 Pro (iOS 26.6.1); update channel option A (source stays private, signed packages in a separate public repo).

1. Spec review

Source: `SDLC Docs Tools software installers and ReadMe/MASTER Specifications/Chedam - Master Specification v1.0.pdf` (41 pages).

The PDF text was extracted with `pdftotext -layout`. The release-plan and architecture pages are images, so they were rendered with PyMuPDF and read as pictures.

P0 requirements: FR-1.01 to 1.10, 1.13, 1.14, 1.15; FR-12.01 to 12.04, 12.08, 12.09.

Done when: 3 devices log in, the hub survives a power pull, an update installs and rolls back. The phase gate adds: memory under 150 MB (NFR-08), a load test (R6), an iOS/Android HTTPS camera test (R5), and a verified backup and restore.

2. Developer PC tools

Already present: Git 2.52.0, VS Code 1.135, Python 3.13, Windows OpenSSH.

Tool	Version	Verified against	Result
Node.js	24.21.0 (LTS Krypton)	nodejs.org SHASUMS256 + OpenJS signature	Match, installed
PocketBase (Windows)	0.40.4	PocketBase <code>checksums.txt</code>	Match
DuckDB CLI	1.5.6	GitHub release digest	Match
Raspberry Pi Imager	2.0.11.1	GitHub digest + Raspberry Pi Ltd signature	Match, installed
DB Browser for SQLite	3.13.1	winget manifest + SignPath signature	Match, installed
mkcert	1.4.4	Scoop manifest (file is unsigned)	Match
GitHub CLI	2.102.0	n/a	Installed, logged in as <code>repatikosamid-prog</code>

Commands used for verification (Git Bash):

```
sha256sum *.msi *.exe *.zip
curl -s https://nodejs.org/dist/v24.21.0/SHASUMS256.txt | grep x64.msi
```

```
powershell Get-AuthenticodeSignature <file>
```

Failed case: `mkcert` not recognized

- Symptom: `mkcert -install` → "The term 'mkcert' is not recognized".
- Cause: the file was still named `mkcert-v1.4.4-windows-amd64.exe`, and `C:\Tools` was not on PATH. PowerShell also never runs a program from the current folder without `.\`.
- Fix: copied `mkcert.exe`, `pocketbase.exe` and `duckdb.exe` to the top of `C:\Tools` and added `C:\Tools` to the user PATH. A new PowerShell window was needed for the change to show.

Then Sreya ran `mkcert -install` (local dev CA added to the Windows trust store) and `gh auth login` (fine-grained token).

The installers folder was added to `.gitignore`. PocketBase had created test data there.

3. Flashing the Pi (first time)

Imager settings: Pi Zero 2 W, Raspberry Pi OS Lite 64-bit, hostname `chedam`, user `chedam`, Wi-Fi 2.4 GHz, country CA, time zone America/Vancouver, SSH public-key only.

Failed case 1: hostname and username

- The hostname was typed as `Chedam` and the username as the GitHub name.
- Fix: lowercase `chedam` for both, so every script and guide matches.

Failed case 2: no SSH key existed

- SSH was set to password authentication, and the PC had no key yet.
- Fix: `ssh-keygen -t ed25519 -C "chedam-dev"`.

Failed case 3: `Permission denied (publickey)`

- Debug output showed the PC offering its key and the Pi refusing it for both usernames:

```
ssh -v -o BatchMode=yes chedam@chedam.local true
```

- Cause: the key written to the card was not the PC's key.
- The "host key differs" warning came from an old `192.168.1.241` entry in `known_hosts`. It was removed with `ssh-keygen -R`.

Failed case 4 (security): private key pasted into Imager

- Imager listed 7 "keys". They were the 7 lines of the **private** key (`BEGIN OPENSSH PRIVATE KEY`).
- Risk: the private key was copied onto the SD card and partly shown in a screenshot.
- Fix: removed all 7 entries, deleted the key pair, generated a new one, and pasted **only** `id_ed25519.pub` (one line, `ssh-ed25519 ... chedam-dev`). Re-flashing overwrote the card.

Failed case 5: `.local` name not resolving from Windows tools

- Git Bash ssh and Windows ssh both failed on `chedam.local`, while `Resolve-DnsName` worked.
- Fix: an SSH alias `chedam` pointing at the IP, in `~/.ssh/config`. The Pi first booted slowly (~5 min), so early checks also failed.

Login worked: `chedam@chedam`, Debian 13 trixie, aarch64, 415 MB RAM, 117 GB card.

Admin access

`sudo` asked for a password, so Sreya ran this once:

```
ssh -t chedam "echo 'chedam ALL=(ALL) NOPASSWD:ALL' | sudo tee /etc/sudoers.d/010-chedam && sudo chmo
```

4. Hub setup (first time, Pi at 192.168.1.241)

OS update (ran in the background, ~15 min):

```
sudo apt-get update && sudo apt-get -y full-upgrade && sudo apt-get -y install chrony sqlite3 ufw
```

PocketBase 0.40.4 linux_arm64 was downloaded on the Pi and checked with `sha256sum -c` against `checksums.txt`.

Files written in the repo:

File	Purpose
hub/scripts/setup-hub.sh	Idempotent setup: packages, service user <code>chedam-hub</code> , <code>/opt/chedam</code> , PocketBase service, Caddy, watchdog, journald, Wi-Fi power save off, boot config, firewall
hub/system/chedam-hub.service	PocketBase on 127.0.0.1:8090, auto-restart, <code>GOMEMLIMIT=100MiB</code> , <code>MemoryMax=220M</code> , hardening
hub/system/Caddyfile	HTTPS for <code>chedam.local</code> and the hub IP, using a per-store CA; plain HTTP only for <code>/ca.crt</code>
hub/system/watchdog.conf	Hardware watchdog 14 s
hub/system/journald.conf	Logs capped at 64 MB (SD card wear)
hub/system/wifi-powersave-off.conf	Lower Wi-Fi latency
hub/system/index.html	Placeholder page "Chedam hub is running"

Run on the Pi:

```
scp -r scripts system chedam:hub/
ssh chedam 'sudo bash ~/hub/scripts/setup-hub.sh'
ssh chedam 'sudo systemctl reboot'
```

Failed case 6: GitHub repo creation

- `gh repo create ... --public` → "Resource not accessible by personal access token".
- Cause: the fine-grained token cannot create repositories. Sreya has to create the public repo on github.com.

Failed case 7: HTTPS tests from the PC returned nothing

- Windows `curl` (Schannel) → "the revocation status is unknown". A private CA has no revocation server.
- Fix for command-line tests: `--ssl-no-revoke`. Browsers are not affected.

Failed case 8: Caddy error at start

- Caddy tried to install its root certificate into the Pi's own trust store and failed ("failed to execute sudo").
- Fix: the `skip_install_trust` global option in the Caddyfile.

Results after reboot

Check	Result
Services	chedam-hub, caddy, chrony, avahi-daemon active
HTTPS	<code>https://chedam.local/api/health</code> and <code>https://192.168.1.241/api/health</code> → 200
Certificate	Valid against "Chedam Store CA - 2026 ECC Root" (valid to 2036); IP cert has IP SAN
Memory	PocketBase 27 MB + Caddy 41 MB; total RAM rose from 415 to 462 MB after headless config
Watchdog	14 s
Temperature	52.6 °C, no throttling

5. Decisions recorded (docs/P0-mini-spec.md)

ID	Decision
DL-21	HTTPS via Caddy with an internal CA created on each hub (one per store); mkcert is dev-PC only
DL-22	Source repo private; signed update packages in a separate public repo
DL-23	RTC deferred for dev; required for the pilot
DL-24	Headless boot config (later changed to a switch, see 2026-10-05)
DL-25	PocketBase extended with JavaScript hooks; no Go build

Findings:

- Android 9 cannot resolve `.local`, so the hub needs a fixed IP.
- Android 9's Chrome no longer gets updates.
- Caddy leaf certificates last 12 hours, so the clock must be correct (RTC for the pilot).